Energy test revision: "I can" checklist

Year 8 SEAL Science · Chapter 7 Energy · Kai's Off-Grid Year

Name: _____ Class: _____ Date: _____

How to use it. Read each line with your book closed. Tick the first box if you can do it now. Revise the unticked lines first, then tick the second box when you can.

Where to revise. Each lesson shows where to look on the Kai site and in Good Science 8.

Lesson 1 (7.0) · Kai's first night: What is energy?	**Now**	**After revising**
I can explain what energy is in my own words.	☐	☐
I can spot where energy is in an everyday situation.	☐	☐
I can tell the difference between energy that is happening now and energy that is stored.	☐	☐
Revise: Lesson 1: Make and Check steps. Textbook: Section 7.1.		

Lesson 2 (7.1) · The watermill: Types of energy, KE and GPE	**Now**	**After revising**
I can sort types of energy into kinetic and potential.	☐	☐
I can calculate kinetic energy with KE = ½mv², squaring the speed first.	☐	☐
I can calculate gravitational potential energy with GPE = mgh.	☐	☐
I can convert grams to kilograms and centimetres to metres before I calculate.	☐	☐
I can use GPE and KE together to work out a speed or a height.	☐	☐
Revise: Lesson 2: Learn and Explore steps (energy simulation and calculation drills). Textbook: Section 7.1.		

Lesson 3 (7.2) · Kai's fire: Heat transfer	**Now**	**After revising**
I can name conduction, convection or radiation for an everyday example.	☐	☐
I can explain conduction using particles that vibrate and bump their neighbours.	☐	☐
I can explain how a convection current forms, including what happens to the density of the fluid.	☐	☐
I can explain why radiation is the only one that works through empty space.	☐	☐
I can use conductors and insulators to explain everyday examples.	☐	☐
Revise: Lesson 3: Learn and Explore steps (heat transfer simulation). Textbook: pp. 170–171.		

Lesson 4 (7.3) · The panel: Transfers, transformations, efficiency, Sankey diagrams	**Now**	**After revising**
I can explain the difference between an energy transfer and an energy transformation.	☐	☐
I can state the law of conservation of energy.	☐	☐
I can calculate efficiency as a percentage, and the energy wasted.	☐	☐
I can draw a Sankey diagram to scale, with every arrow labelled.	☐	☐
I can read a Sankey diagram to find useful and wasted energy.	☐	☐
Revise: Lesson 4: Learn and Explore steps (swing and Sankey simulation). Textbook: pp. 172–177.		

Lesson 5 (7.4) · Fire knowledge: First Nations energy and fire	Now	After revising
I can describe how First Nations Peoples start fire without matches.	☐	☐
I can draw an energy flowchart for a fire-starting method, labelling transfers and transformations.	☐	☐
I can explain why fire has mattered to First Nations Peoples for tens of thousands of years.	☐	☐
Revise: Lesson 5: Learn and Explore steps (fire-starting simulation). Textbook: pp. 178–179.		

Lesson 6 (7.5) · Wiring up: Appliances, star ratings and running cost	Now	After revising
I can explain the difference between power (W) and energy (Wh or kWh).	☐	☐
I can calculate energy with energy (Wh) = power (W) × time (h).	☐	☐
I can convert Wh to kWh, and cents to dollars.	☐	☐
I can calculate running cost with a tariff.	☐	☐
I can read every number on an energy rating label and say what each one tells me.	☐	☐
Revise: Lesson 6: Learn and Explore steps (appliance simulation). Textbook: pp. 180–181.		

Lesson 7 (7.6) · The cabin: Household energy and design	Now	After revising
I can explain what an R-value tells me about insulation.	☐	☐
I can name the type of heat transfer each home upgrade reduces.	☐	☐
I can describe passive design features that keep a home comfortable without appliances.	☐	☐
I can explain how solar hot water and other energy-saving technologies work.	☐	☐
I can use data, including cost, to choose the best upgrade.	☐	☐
Revise: Lesson 7: Learn and Explore steps (heat leaks and passive design simulation). Textbook: pp. 182–185.		

Lesson 8 (7.7) · Insulation prac: Insulation and investigation skills	Now	After revising
I can describe bulk, reflective and vacuum insulation.	☐	☐
I can name the independent, dependent and controlled variables in an investigation.	☐	☐
I can describe a trend in data, quoting numbers with units.	☐	☐
I can spot a result that doesn't fit the pattern, and suggest what went wrong.	☐	☐
I can explain the difference between reliable and valid.	☐	☐
Revise: Lesson 8: Learn step and your practical report; Lesson 10 evaluation station. Textbook: p. 188.		

Lesson 9 (7.8) · Energy budget: Home energy audit	Now	After revising
I can explain what an energy audit looks at.	☐	☐
I can read a pie chart or table of household energy use.	☐	☐
I can check whether energy supply is enough to meet demand.	☐	☐

I can explain how behaviour, design and appliance choice change energy use.	☐	☐
Revise: Lesson 9: Learn and Explore steps (home audit simulation). Textbook: pp. 190–191.		

Lesson 10 (7.9) · Revision: Class revision	**Now**	**After revising**
I can answer mixed questions from every lesson without my notes.	☐	☐
I can explain a whole energy system using conservation of energy and efficiency.	☐	☐
I can name my two weakest topics and write a revision plan as actions.	☐	☐
Revise: Lesson 10: mixed revision quiz, 'Am I ready?' list and evaluation station. Textbook: Whole chapter.		

My revision plan

Write three actions for your weakest topics, like "Redo the Sankey drills in Lesson 4", not "study efficiency".

1. _____

2. _____

3. _____